

BENGALURU, India

WMO: 432950

Lat: 12.967N

Lon: 77.583E

Elev: 3022

StdP: 13.16

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	59.9	61.2	34.9	33.2	83.5	39.0	39.1	82.0	9.9	68.7	9.0	70.4	2.4	90	0.226

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	18.5	93.8	68.0	92.4	67.8	90.8	67.8	74.6	84.4	73.8	83.4	73.0	82.4	2.8	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.1	133.0	78.3	71.2	128.9	77.4	70.6	126.1	76.7	40.5	84.5	39.6	83.8	38.9	82.5	82.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
11.5	9.6	8.0	DB	56.6	96.9	1.6	1.7	55.5	98.2	54.5	99.2	53.7	100.2	52.5	101.5
			WB	51.3	77.9	1.8	2.1	50.0	79.4	48.9	80.7	47.9	81.9	46.6	83.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	75.9	71.8	75.6	80.1	82.3	80.6	76.7	75.0	74.7	75.2	74.8	72.6	70.9
	DBStd	4.08	2.08	2.68	2.75	2.62	2.85	2.30	1.78	1.66	1.79	2.02	2.19	2.14
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	9439	676	716	933	968	948	802	776	766	757	769	679	649
	CDD65	3964	211	296	468	518	483	352	311	301	307	304	229	184
	CDH74	28218	1186	2401	4778	5566	4585	2260	1448	1300	1585	1470	914	725
	CDH80	9268	162	816	2178	2589	1940	582	223	150	256	238	89	45
(14) Wind	WSAvg	3.4	2.8	2.7	2.5	2.4	3.4	5.5	5.6	4.6	3.4	2.3	2.5	2.8
(15) Precipitation	PrecAvg	39.9	0.1	0.2	0.5	2.2	4.7	3.7	4.3	5.7	8.5	6.5	2.2	0.8
	PrecMax	64.6	0.9	2.3	4.0	12.8	9.5	9.0	11.3	15.2	20.4	23.9	9.3	4.7
	PrecMin	25.4	0.0	0.0	0.0	0.0	0.0	0.3	0.7	1.8	1.3	0.1	0.0	0.0
	PrecStd	8.7	0.2	0.5	0.9	2.4	2.0	2.3	2.8	3.3	4.5	4.1	2.2	1.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	85.5	91.6	94.5	96.0	95.0	90.4	86.1	85.1	86.2	86.1	84.6	83.3
		MCWB	62.9	63.3	65.2	68.7	69.8	72.5	70.1	70.6	69.9	67.1	65.1	63.0
	2%	DB	83.4	88.9	92.9	94.3	93.1	87.2	84.3	83.2	84.4	84.2	82.4	81.2
		MCWB	62.8	63.3	65.2	68.4	70.3	72.1	70.6	70.6	69.8	67.9	65.9	63.6
	5%	DB	81.8	87.0	91.4	92.8	91.2	85.3	82.5	81.7	82.9	82.6	80.8	79.7
		MCWB	62.3	62.9	65.1	68.2	70.9	71.6	70.4	70.4	69.9	68.5	66.3	63.6
	10%	DB	80.2	85.1	89.8	91.0	89.1	83.5	80.8	80.2	81.3	81.1	79.2	78.2
		MCWB	62.3	62.9	64.6	68.5	71.3	71.2	70.1	70.2	69.8	68.9	66.2	63.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	69.3	70.7	73.1	75.2	76.0	75.6	73.6	73.7	73.7	73.4	72.0	70.2
		MCDB	78.0	81.0	84.4	86.2	86.5	84.8	80.9	80.8	80.6	80.2	77.9	76.0
	2%	WB	67.8	69.3	71.7	73.8	74.7	74.1	72.4	72.4	72.6	72.3	70.7	69.0
		MCDB	76.3	79.3	82.0	84.8	85.1	83.3	80.0	79.4	79.5	78.7	76.3	74.5
	5%	WB	66.8	68.3	70.6	72.8	73.9	73.1	71.6	71.6	71.8	71.6	69.9	68.0
		MCDB	75.2	77.9	81.3	83.7	84.3	81.9	79.2	78.5	78.6	77.7	75.5	73.4
	10%	WB	65.8	67.2	69.6	72.0	73.2	72.1	70.9	71.0	71.1	70.9	69.1	67.0
		MCDB	74.0	76.8	80.7	82.6	83.4	80.3	78.0	77.4	77.6	76.6	74.5	72.6
(35) Mean Daily Temperature Range	5% DB	MDBR	18.5	20.3	20.2	18.5	17.3	13.6	12.2	11.7	12.7	12.8	13.6	15.9
		MCDBR	20.4	22.2	21.2	19.9	19.8	16.2	14.7	13.7	15.0	15.3	16.7	18.2
		MCWBR	5.4	6.0	7.1	7.2	5.9	4.9	4.5	4.3	4.4	4.6	5.0	5.2
	5% WB	MCDBR	15.9	17.4	18.0	18.1	17.4	14.4	12.8	12.1	12.4	11.9	11.4	12.4
		MCWBR	5.6	5.9	7.2	7.0	6.5	5.6	4.9	4.7	4.7	4.7	4.6	5.2
(40) Clear-Sky Solar Irradiance	taub		0.383	0.404	0.466	0.508	0.522	0.454	0.466	0.454	0.440	0.472	0.413	0.389
	taud		2.306	2.251	2.084	2.013	2.003	2.254	2.210	2.250	2.284	2.169	2.313	2.332
	Ebn at Noon		290	288	272	259	250	265	263	269	275	264	279	285
	Edh at Noon		41	44	54	57	57	44	46	45	43	48	40	39
(44) All-Sky Solar Radiation	RadAvg		1715	1962	2074	2084	1986	1660	1479	1500	1636	1572	1465	1504
	RadStd		77	81	71	68	97	122	92	92	92	111	133	106

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.65	+0.70	N/A	+0.90	N/A	N/A	N/A	+308	+310
(47) Regional (3 neighbors)		+0.58	N/A	N/A	+0.94	N/A	N/A	N/A	+308	+310

Nomenclature: See separate page